mind of a positivist. To him the stars are nothing more than complexes of optical sensations; he considers everything else as just a useful but basically arbitrary and unessential trimming.

But let us now leave Positivism and continue with our own train of thought. The law of the conservation of energy is, after all, not the only law of nature, it is just one among many. While it is true that it is universally valid, it is still not sufficient for predicting every detail of a natural process, since it leaves an endless number of possibilities still open.

But there is another, far broader law, which has the property of giving a specific, unequivocal answer to each and every sensible question concerning the course of a natural process; as far as we can see, this law—like the law of the conservation of energy—possesses an exact validity even for the most modern parts of physics. But what we must regard as the greatest wonder of all, is the fact that the most adequate formulation of this law creates the impression in every unbiased mind that nature is ruled by a rational, purposive will.

Let me illustrate this by a specific example. It is a well known fact that when a ray of light strikes the surface of a transparent body obliquely, such as the surface of water, it is deflected from its original direction after penetrating the surface. The explanation of this deflection is that light travels more slowly in water than in air. Such deflection, or “refraction,” occurs also in the atmosphere, because light travels more slowly in the lower, denser strata of the asmosphere than in its higher layers. Now then, when a light ray emitted by a star reaches the eyes of an observer, its path will show a more or less complicated curve, due to the different degrees of refraction in the various atmospheric layers (unless the star happens to be exactly in the zenith). This curve is fully determined by the following simple law: Out of all the possible paths leading from the star to the eye of the observer, light will always follow the one which it can cover in the shortest time, allowance being made for the differences in its velocity in different atmospheric layers. Thus, the photons which constitute a ray of light behave like intelligent beings: Out of all the possible curves they always select the one which will take them most quickly to their goal.

This principle permits a large-scale generalization. According to all that we know about the laws relating to the processes taking place in any physical structure, we can characterize in all its details the course of each process by the principle that among all the conceivable processes which can change the state of a given physical structure into another state during a certain time interval, the process which actually takes place is always one for which the integral over that time interval of a certain magnitude, the so-called Lagrange function, has the smallest value. Therefore, if we know the value of the Lagrange function, we can fully specify the course of the process actually